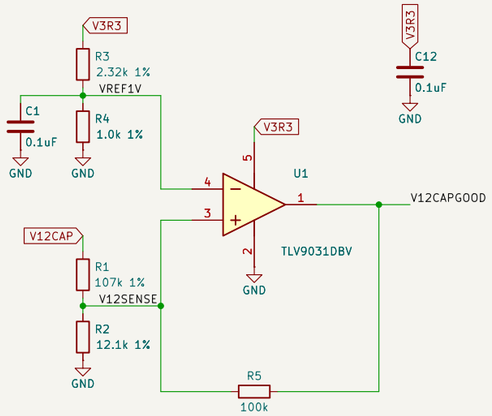
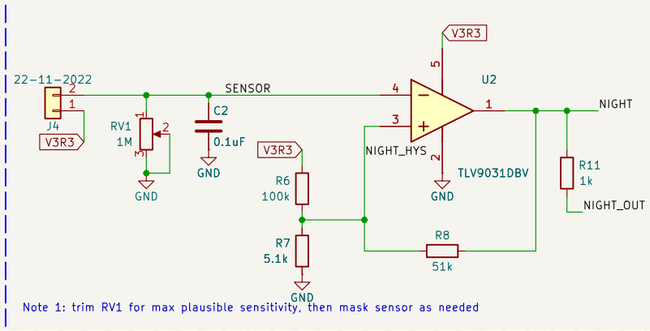
V12CAP Monitor (U1)

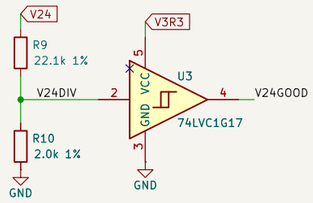


Night Sensor (U2, J4)

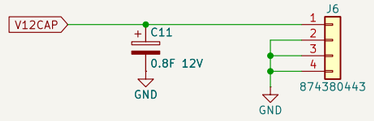


Note 1: trim RV1 for max plausible sensitivity, then mask sensor as needed

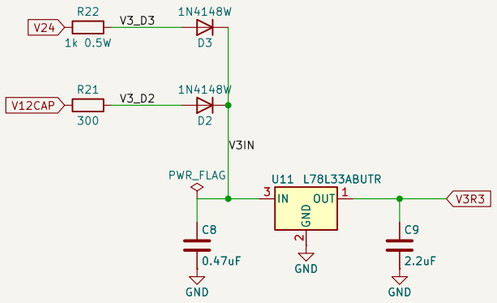
V24 Detect (U3)



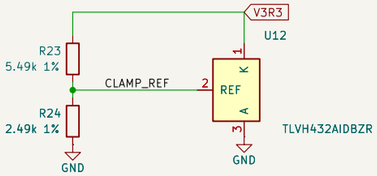
Backup Capacitor (C11, J6)



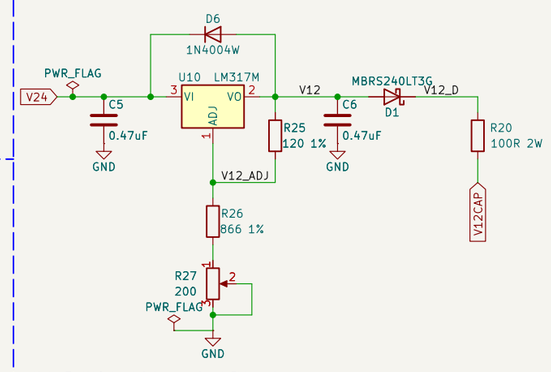
3V3 Regulator (U11)



3V3 Clamp (U12)

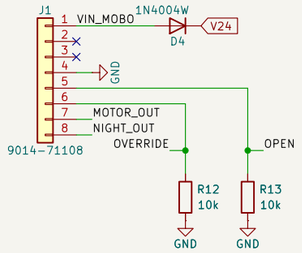


12V Regulator (U10)

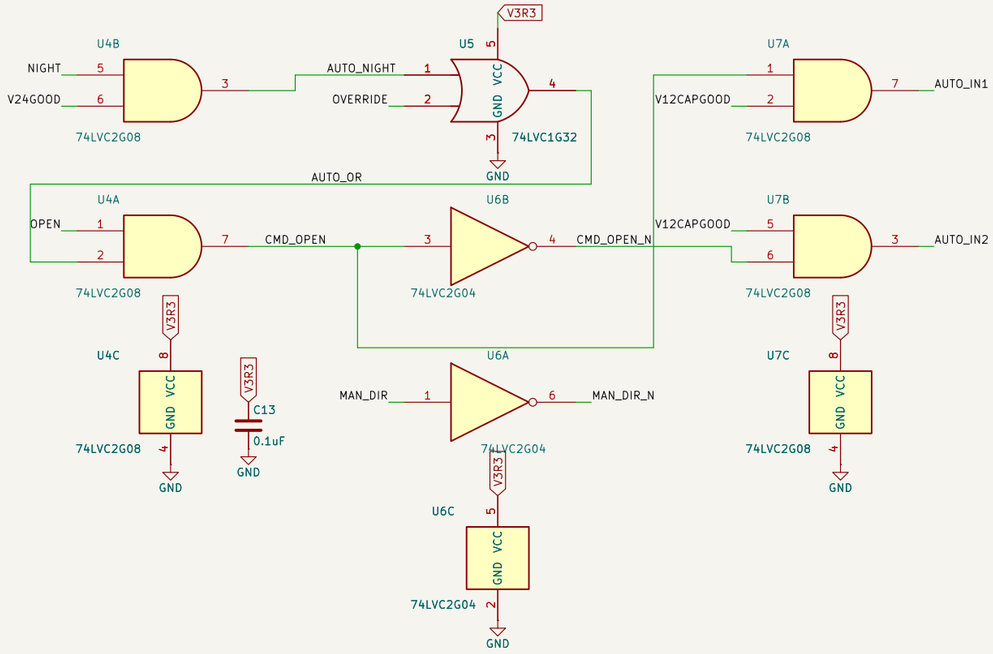


Note 3: U10 heatsink area for 1.5W peak:
suggest min 1 cm^2, prefer 12mm x 12mm or equiv.
pin 2 may be absent

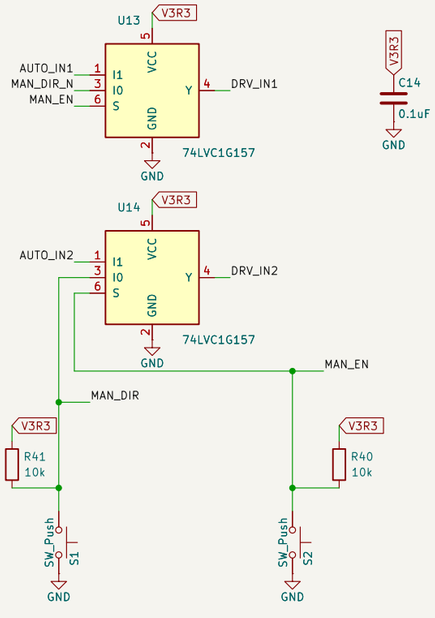
MOBO Connector (J1)



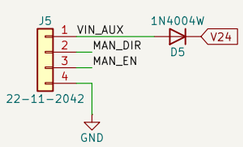
Control Logic (U4–U7)



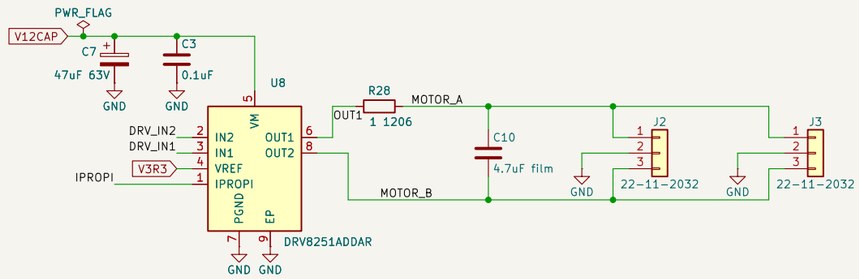
Manual/Auto Select (U13, U14)



Aux Power / Manual (J5)



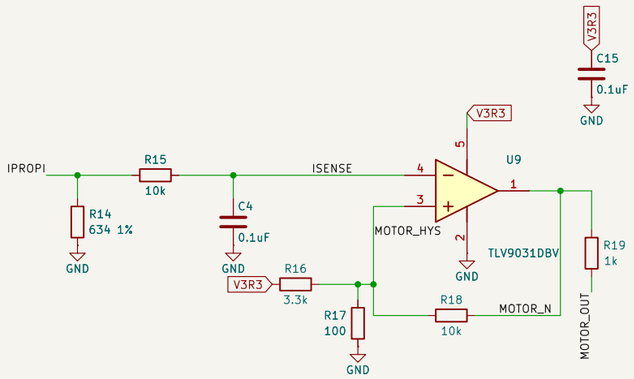
Motor Driver (U8, J2, J3)



DRV8251A truth table

IN1	IN2	OUT1	OUT2
0	0	Hi-Z	Hi-Z
0	1	L	H
1	0	H	L
1	1	L	L

Current Sense (U9)



Note 2: IPROPI Iout = 1.58 mA/A; node internally clamped to V<=Vref;
motor current limits when V=Vref, here Ilim=3.3A

Sheet: /
File: shuttercontrol.kicad_sch

Title: New Shutter Controller

Size: A3	Date: 2026-11-02	Rev: v.02j
KiCad E.D.A. 10.0.7		Id: 1/1